



The Changing Cost of Homeownership in the U.S.

Stat Squad

Presenters:

Emily Aguilar (emilyo2), Yasmine Chatila (ychati2), Zahabiya Lokhandwala (zlokh2),
Tasneem Mubashshira (tmubas2), Angel Corral-Esparza (acorr24)

<https://github.com/uic-cs418/group-project-stat-squad.git>

PROBLEM STATEMENT

Big Idea

- Why Is Buying a Home in U.S. Getting Harder?

What Led Us To This Issue

- We noticed growing concerns about income not keeping up with housing costs in U.S.

Why This Is Important

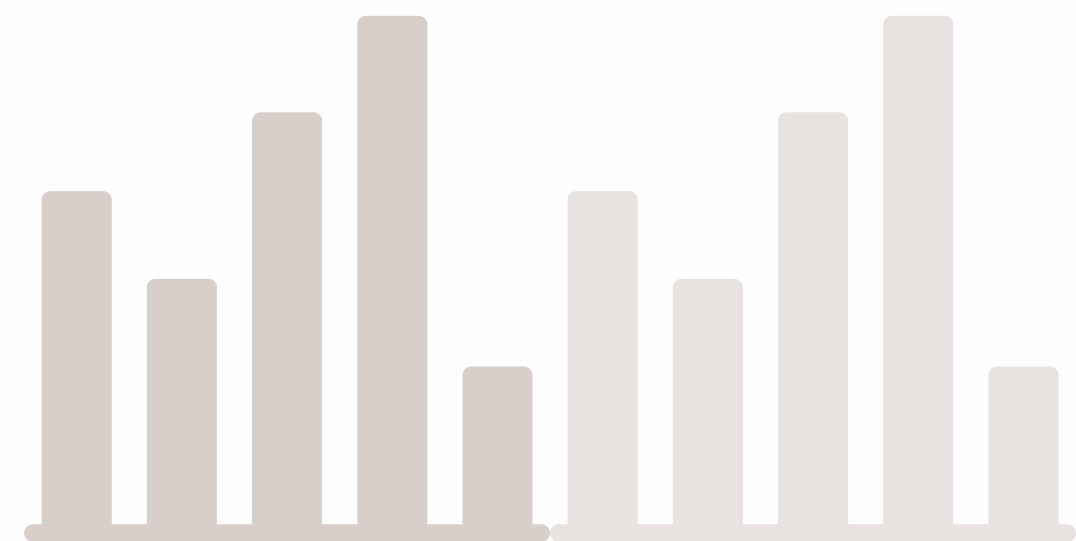
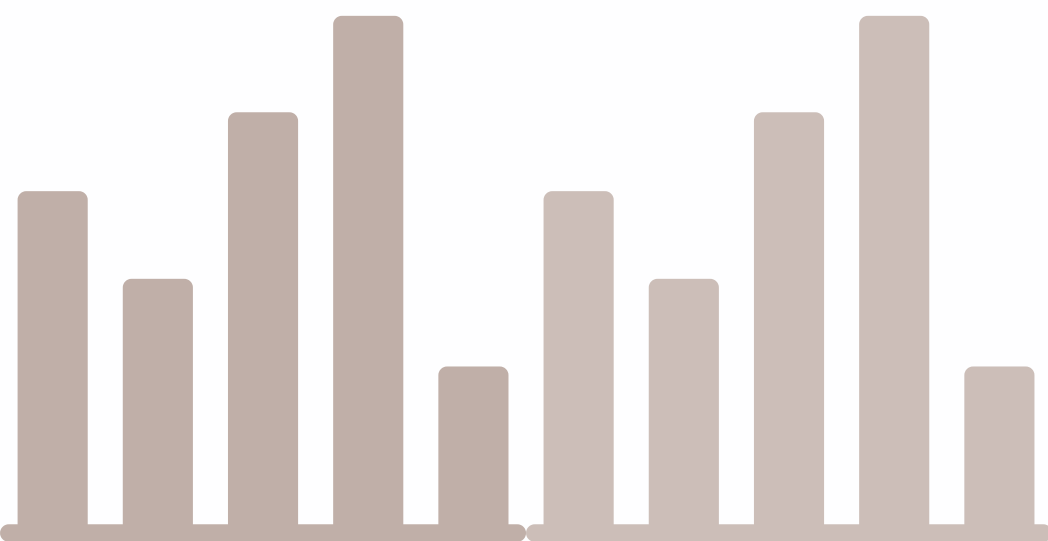
- The widening affordability gap makes homeownership difficult, especially for first-time buyers.

Goal

- By better understanding the affordability gap, we aim to provide insights that could help future homebuyers.



DATASETS DESCRIPTION



Zillow Median Home Values Dataset

- Time series tracking monthly median home values across U.S. regions from **January 2000 to January 2025**
- **Key Features:**
 - SizeRank: Ranking of regions by size.
 - RegionName: Name of the location (e.g., *New York, NY*)
 - StateName: State abbreviation (e.g., *NY, IL*)

Zillow Dataset Sample

	RegionName	StateName	SizeRank	2008-01-31	2008-02-29
0	United States	NaN	0	199183.742747	198013.493044
1	New York, NY	NY	1	450391.791124	449058.133849
2	Los Angeles, CA	CA	2	531251.501468	523160.855250
3	Chicago, IL	IL	3	244764.138456	242664.462515
4	Dallas, TX	TX	4	154724.964939	153717.900038



FRED Annual Real Median Household Income by State Dataset

- Tracks real median household income by state from **1984 to 2023**.
- **Key Features:**
 - State names
 - Annual real median income values

FRED Dataset Sample

	State	2008	2009	2010
0	The United States	68,780	68,340	66,730
1	Alabama	60,810	54,890	55,440
2	Alaska	87,500	84,570	78,340
3	Arizona	64,150	62,790	63,510
4	Arkansas	54,130	50,160	52,260



EXPLORATORY DATA ANALYSIS

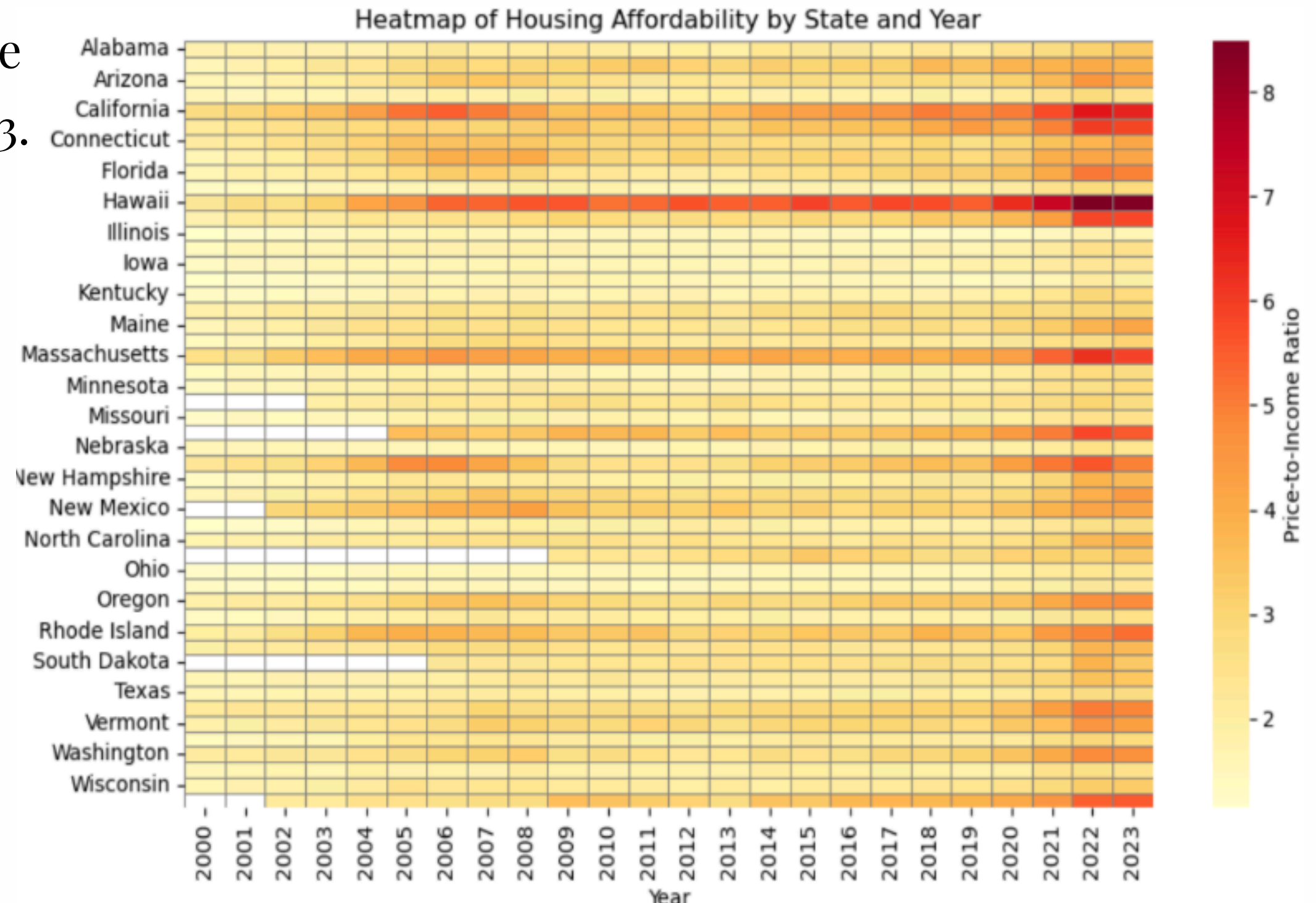


Housing Affordability By State & Year

- This heat map shows the price-to-income ratio for each U.S. state from 2000 to 2023.
- It measures housing affordability
- Price to Income Ratio = Median House Price
- Each row = one state
- Each column = one year

Median House Income Color scale:

- Light yellow = More affordable
- Dark red = Less affordable

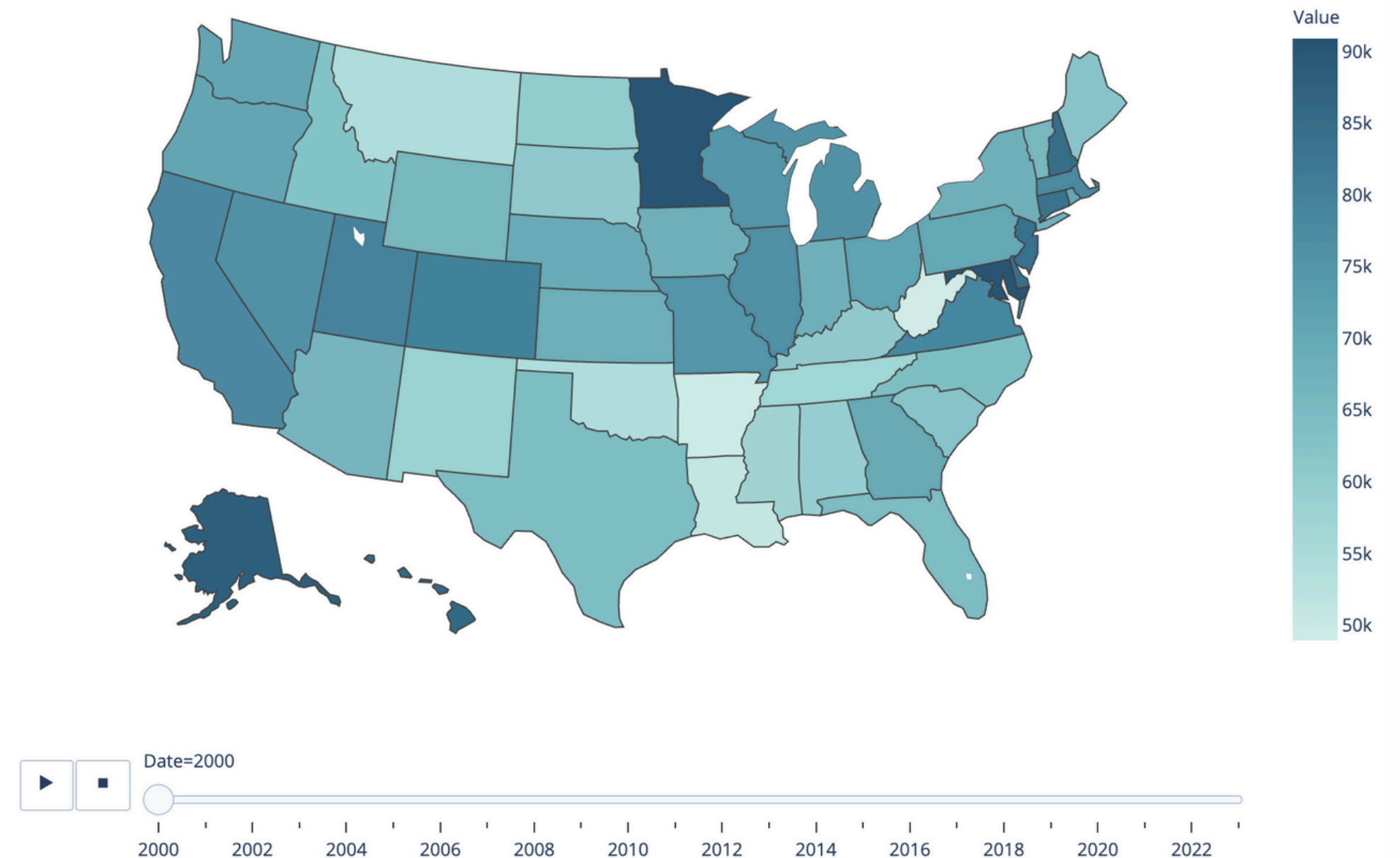


This visualization allows us to compare affordability over time and across regions.

Comparison Of Average Income Throughout The Year

- This choropleth map displays the average income of each state each year
- Through its animation users have the ability to visualize changes of income.
- Darker color intensity represents higher average incomes, typically ranging from \$75K to over \$90K, while lighter shades indicate lower average incomes, generally below \$65K

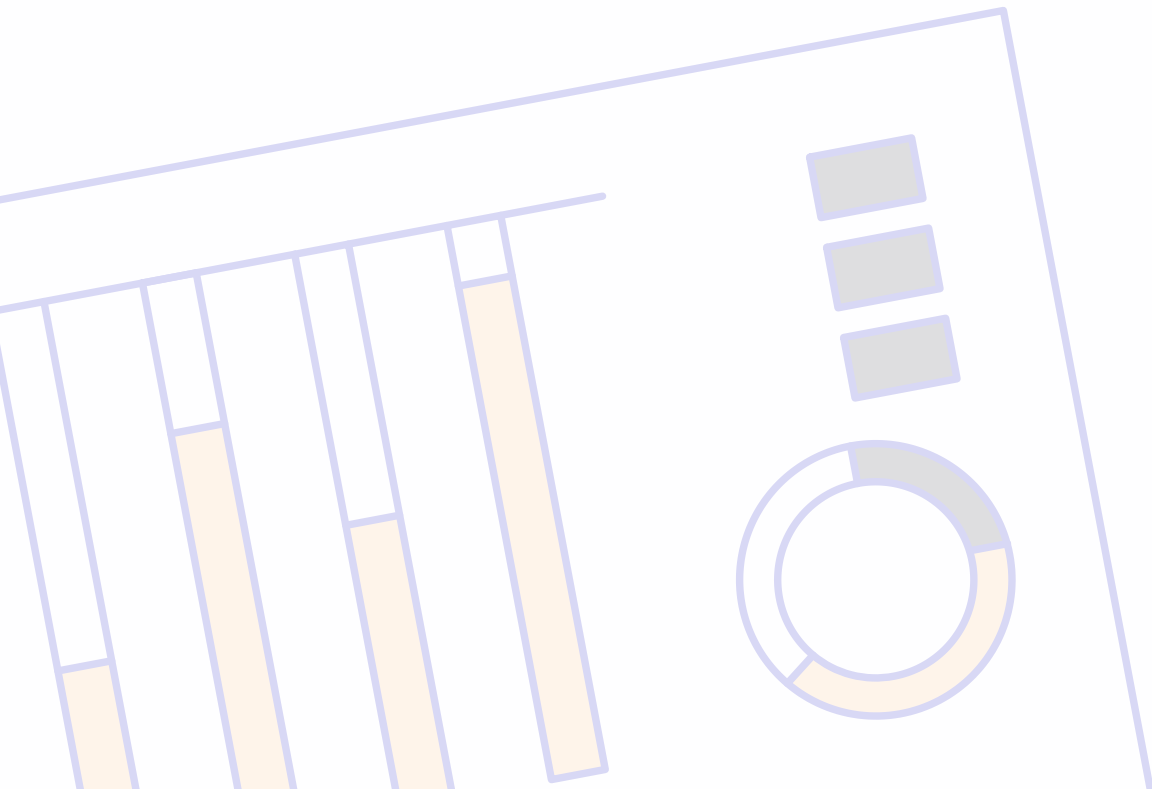
Average Income by Year



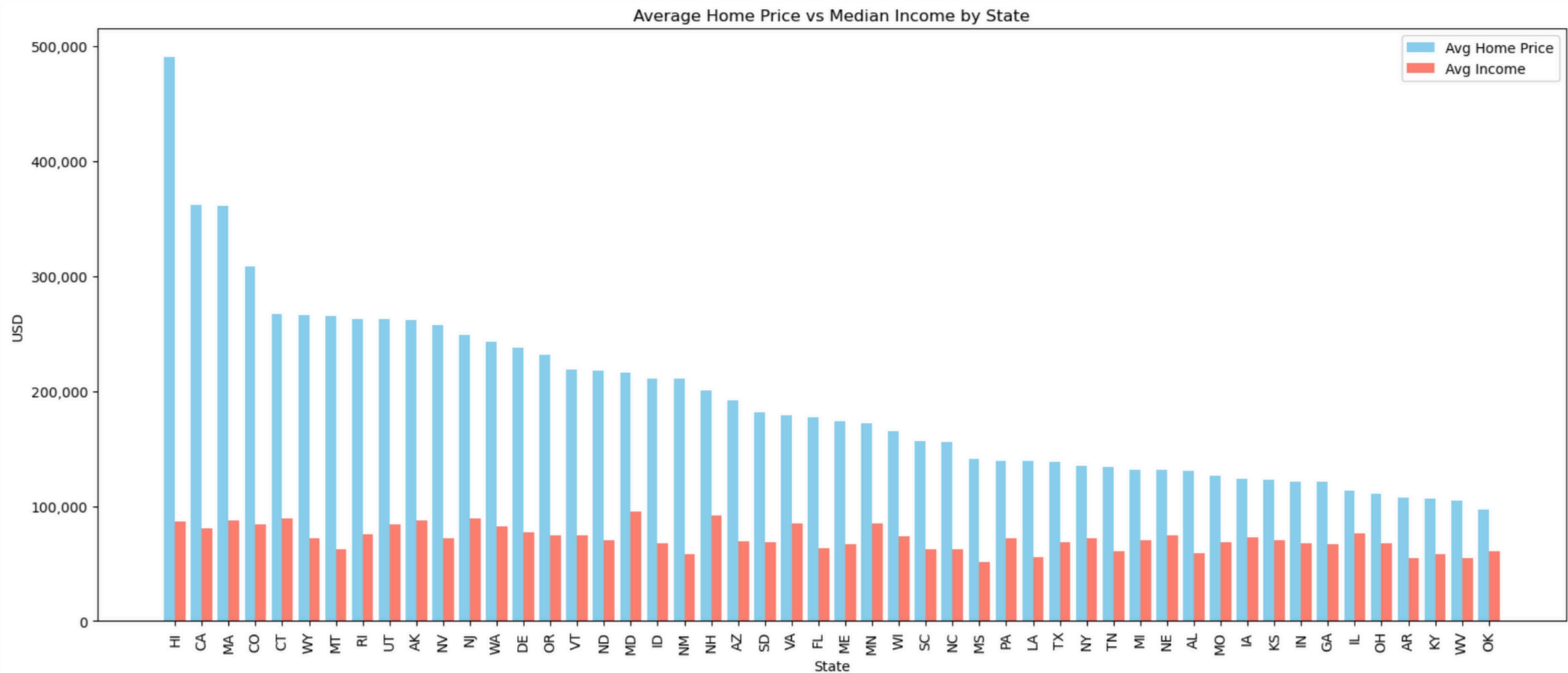
This visualization allows us to compare average income throughout the year



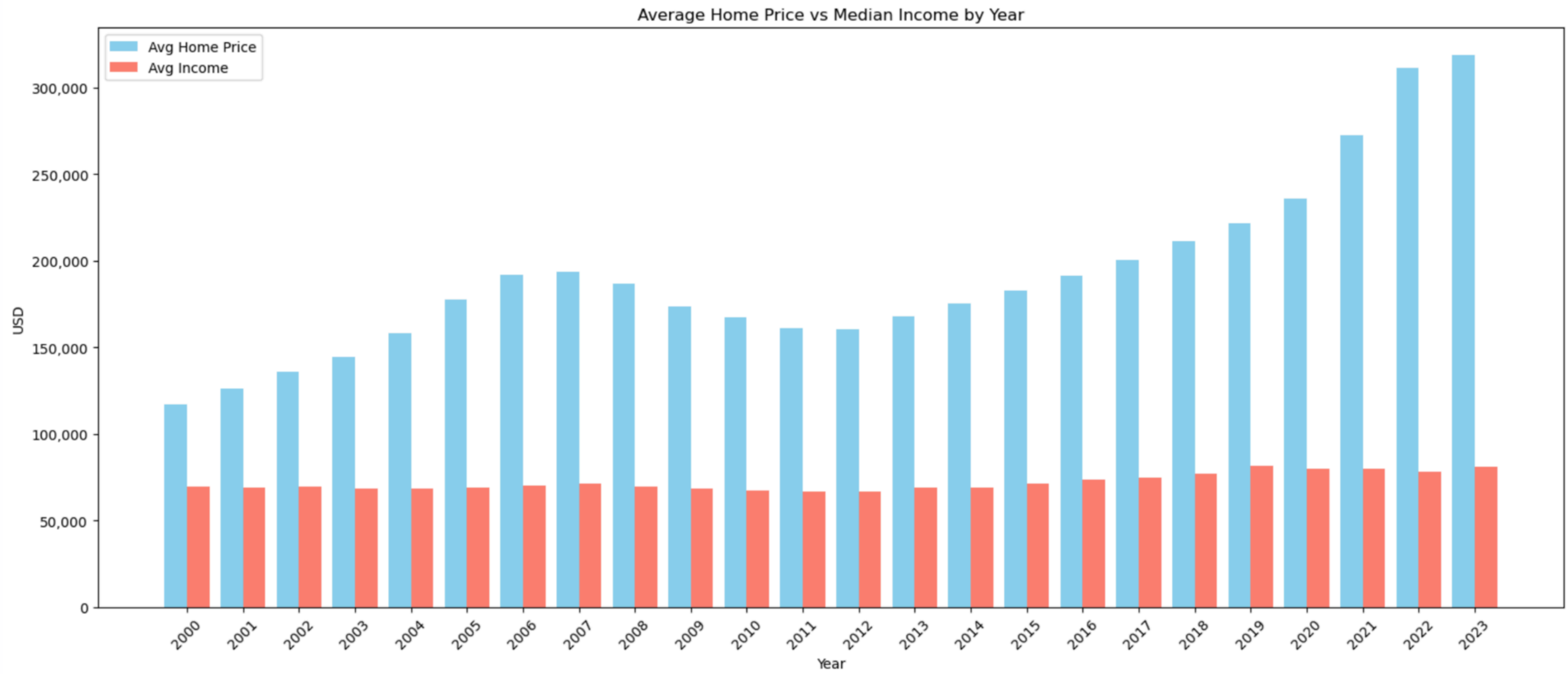
VISUALIZATIONS



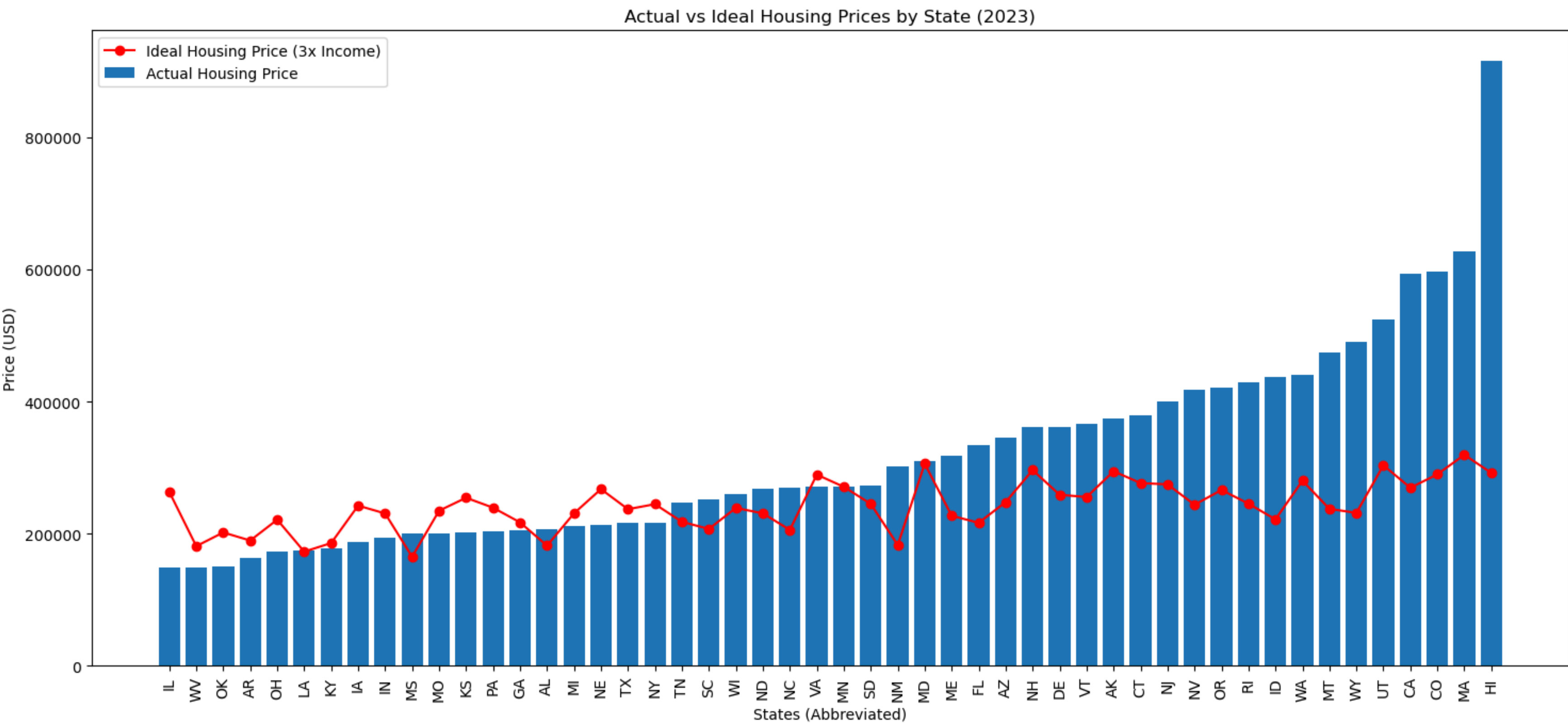
Comparison of Average Home Prices & Median Incomes Across States



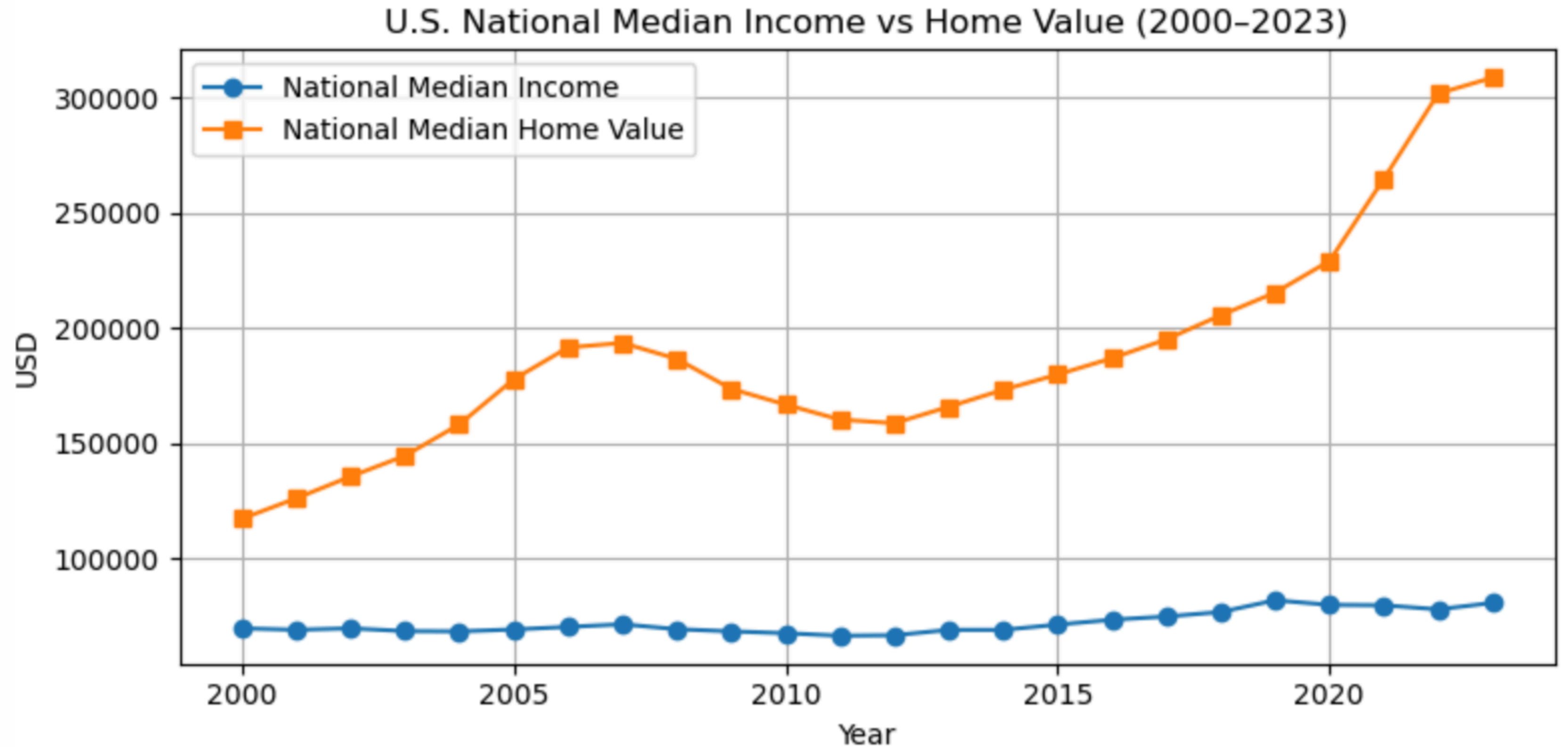
Changes in Average Home Prices & Median Incomes From 2000 To 2023



The Gap Between Ideal and Actual Housing Prices Across States (2023)



Why Homeownership is Slipping Away: U.S. Income vs Housing Costs (2000–2023)



MACHINE LEARNING



K-Means Clustering: Understanding State-Level Housing Affordability

Features Used:

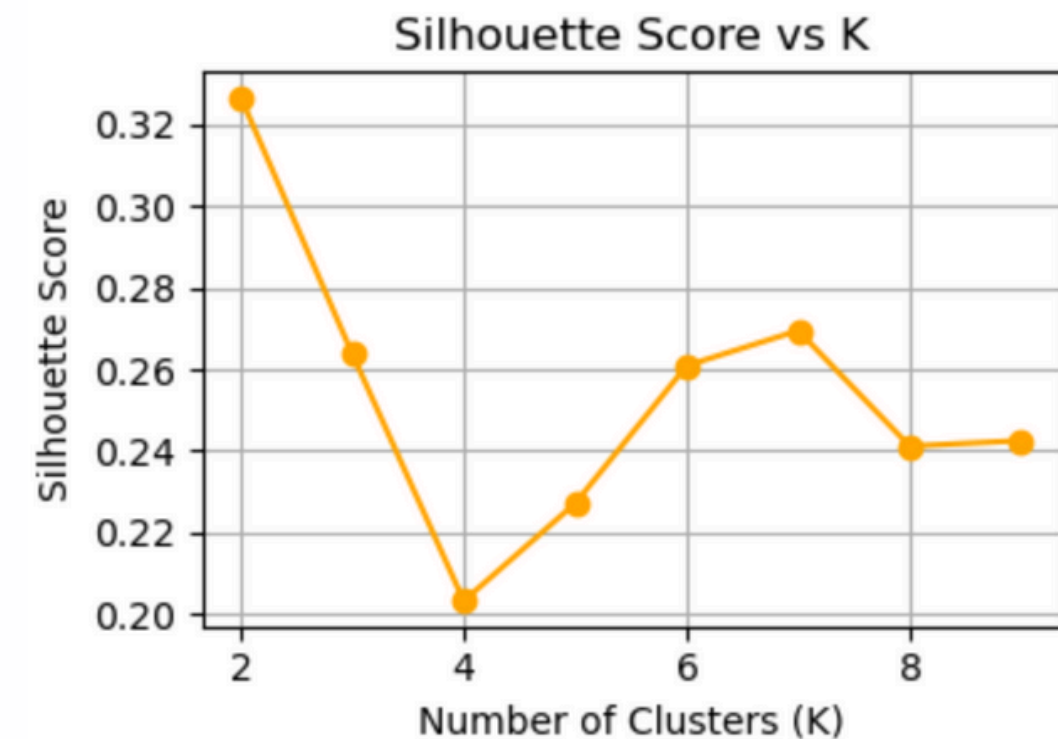
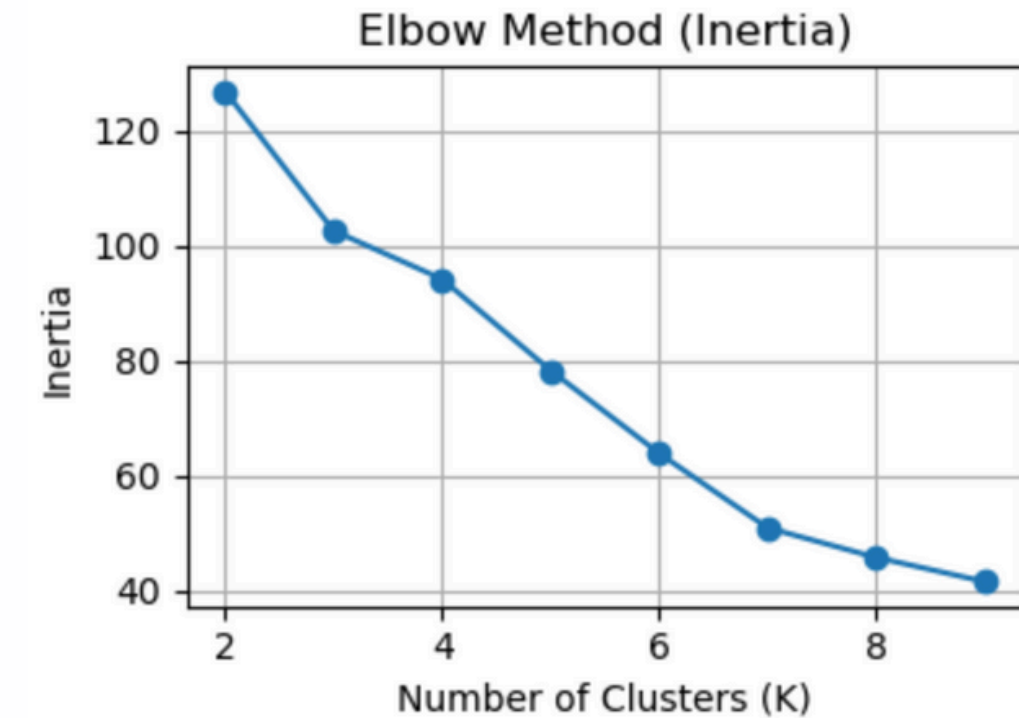
- Latest Median Home Value (2023)
- Home Value Growth Rate (2000–2023)
- Latest Median Income (2023)
- Income Growth Rate (2000–2023)

K = 3 (From Inertia and Silhouette Score)

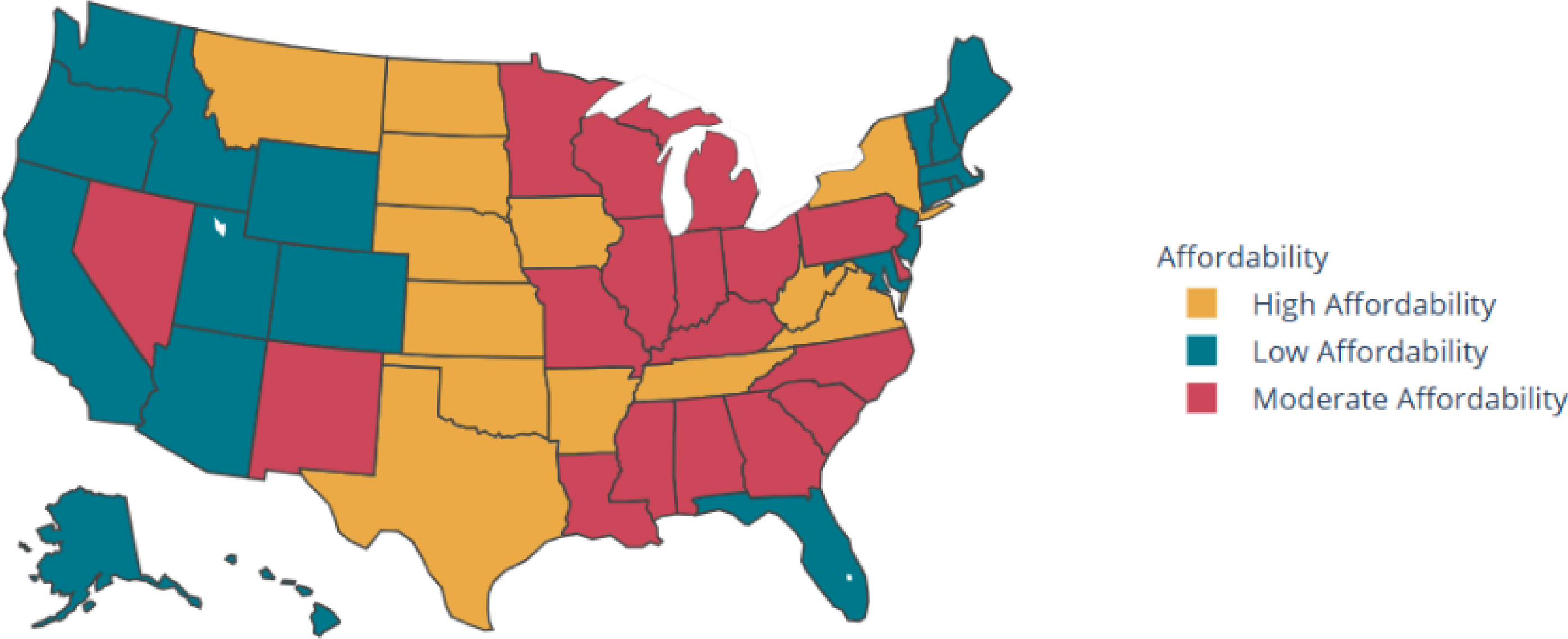
- High Affordability
- Moderate Affordability
- Low Affordability

Baseline Model: Random Clustering

- Silhouette Score (Random Clustering): -0.045
- Silhouette Score (K-Means Clustering): 0.264



The Affordability Divide: Where Americans Can Still Afford a Home

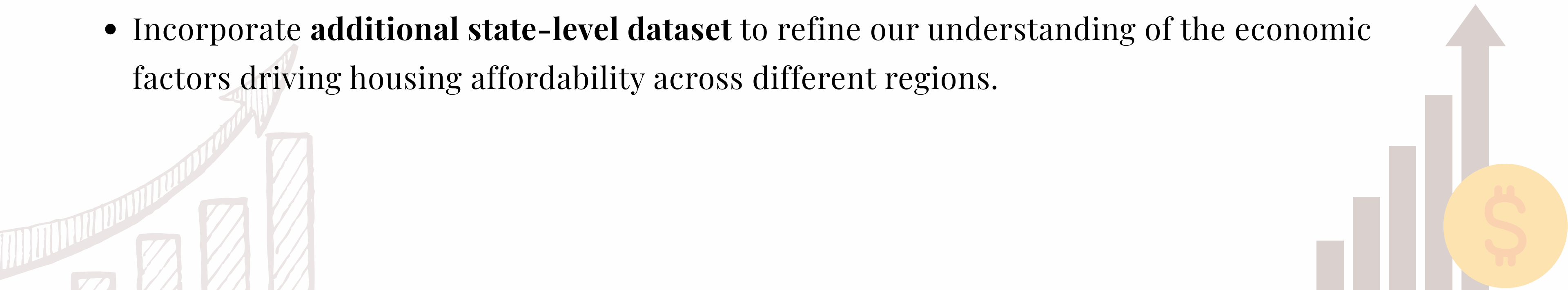


CONCLUSION

- These Data-driven insights emphasize the growing affordability crisis and regional economic disparities across the U.S.
- Understanding these trends is crucial for policymakers, buyers, and anyone concerned about the future of affordable housing in America.

FUTURE WORK

- Analyze monthly housing trends to determine the **best time of year to buy a home** based on historical price patterns.
- Incorporate **additional state-level dataset** to refine our understanding of the economic factors driving housing affordability across different regions.



THANK YOU